

Multivariate Time-Series Prediction Using Deep Learning

Appliance Energy Consumption Forecasting Using Deep Learning Models

Assessment Report

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1. Introduction

Energy consumption forecasting is an important problem in smart home and energy management systems. Accurate prediction of appliance energy usage helps optimize electricity consumption, reduce operational costs, and improve energy efficiency.

The objective of this project was to develop a deep learning-based time-series forecasting system capable of predicting household appliance energy consumption using environmental sensor data and historical appliance usage patterns.

Implemented deep learning architectures:

- LSTM
- GRU
- BiLSTM
- CNN-LSTM
- Hybrid LSTM-GRU

The project focused on data preprocessing, feature engineering, sequential data preparation, model training, optimization, and evaluation.

2. Dataset Insights

The dataset contains appliance energy consumption measurements collected from a smart home environment.

Included variables:

- Appliance energy usage
- Indoor temperatures
- Humidity measurements
- Weather conditions
- Lighting information
- Wind speed and outdoor conditions
- Time-based information

Target Variable:

Appliances

3. Preprocessing

Exploratory Data Analysis identified strong temporal patterns and feature correlations.

Key Findings:

- Energy consumption showed right-skewed distributions.
- Historical lag features strongly correlated with future energy consumption.
- Weakly correlated features were removed to reduce noise.

Preprocessing steps:

- Missing value removal using `df.dropna()`
- Min-Max feature scaling
- 80/20 temporal train-test split

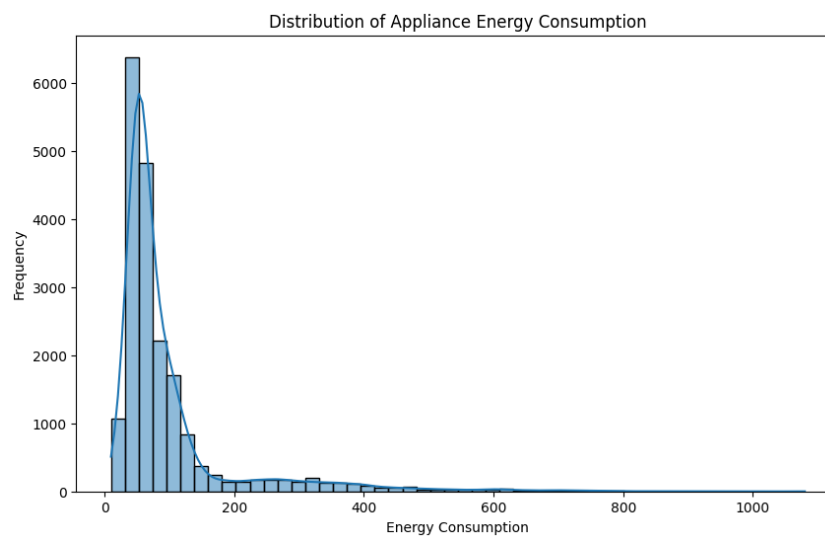


figure 1

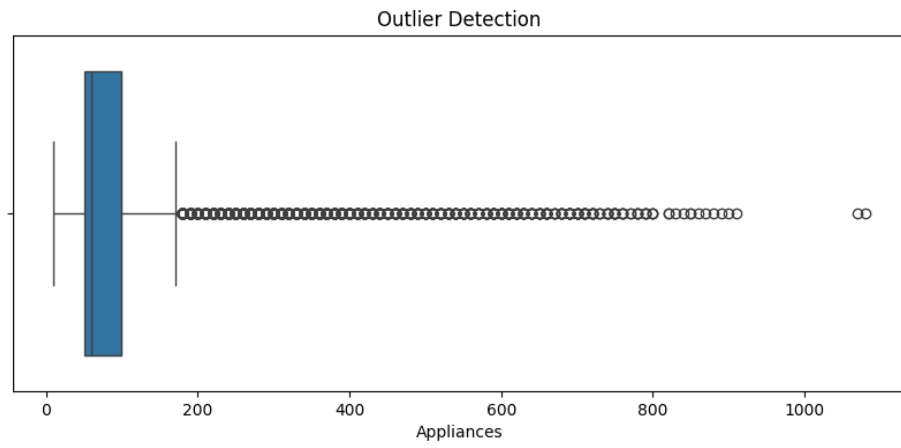


figure 2

4. Feature Engineering

Feature engineering significantly improved model performance.

Created Features:

- Lag Features: lag_1, lag_6, lag_12, lag_24
- Rolling Features: rolling_mean_12, rolling_mean_24, rolling_std_24
- Time-Based Features: hour, lights, day_of_week
- Interaction Features: temp_humidity, indoor_outdoor_temp_difference

Final selected features improved generalization and reduced model complexity.

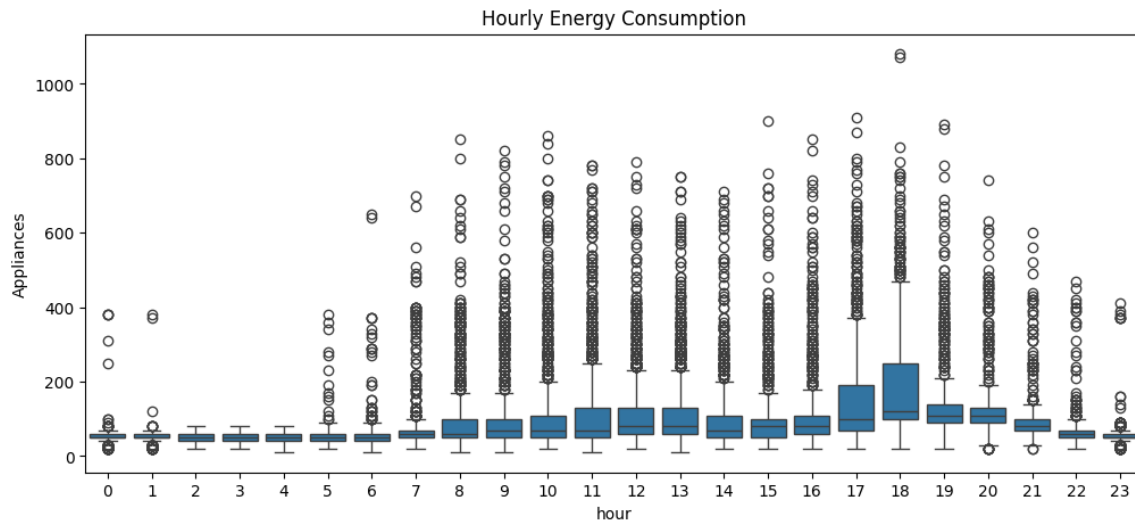


Figure 3

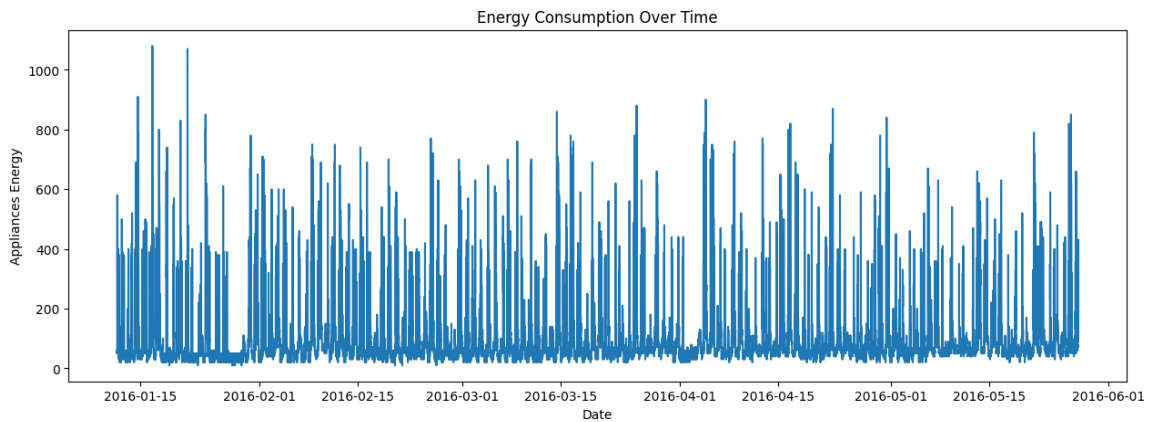


Figure 4

5. Model Design

Multiple machine learning and deep learning architectures were evaluated.

Baseline Models:

- Linear Regression
- Random Forest Regressor

Deep Learning Models:

- LSTM
- GRU
- BiLSTM
- CNN-LSTM
- Hybrid LSTM-GRU

LSTM:

Used stacked recurrent layers to capture long-term dependencies.

GRU:

Implemented as a computationally efficient alternative with fewer parameters.

BiLSTM:

Processed sequences in both directions for improved contextual learning.

CNN-LSTM:

Combined convolutional feature extraction with temporal sequence learning.

Hybrid LSTM-GRU:

Combined LSTM memory capability with GRU efficiency.

Optimizer:

Adam

6. Results

Final Model Performance

Linear Regression:

MAE = 30.11

$R^2 = 0.35$

LSTM:

MAE = 35.75

RMSE = 65.22

$R^2 = 0.448$

GRU:

MAE = 30.51

RMSE = 63.74

$R^2 = 0.473$

BiLSTM:

MAE = 30.15

RMSE = 61.84

$R^2 = 0.5268$

CNN-LSTM:

MAE = 29.42

RMSE = 60.91

$R^2 = 0.5331$

Hybrid LSTM-GRU:

MAE = 27.84

RMSE = 59.67

$R^2 = 0.5474$

Optimized GRU:

MAE = 26.91

RMSE = 59.08

$R^2 = 0.5514$

The optimized GRU model achieved the best forecasting performance and demonstrated improved generalization, stable convergence, and reduced computational complexity.

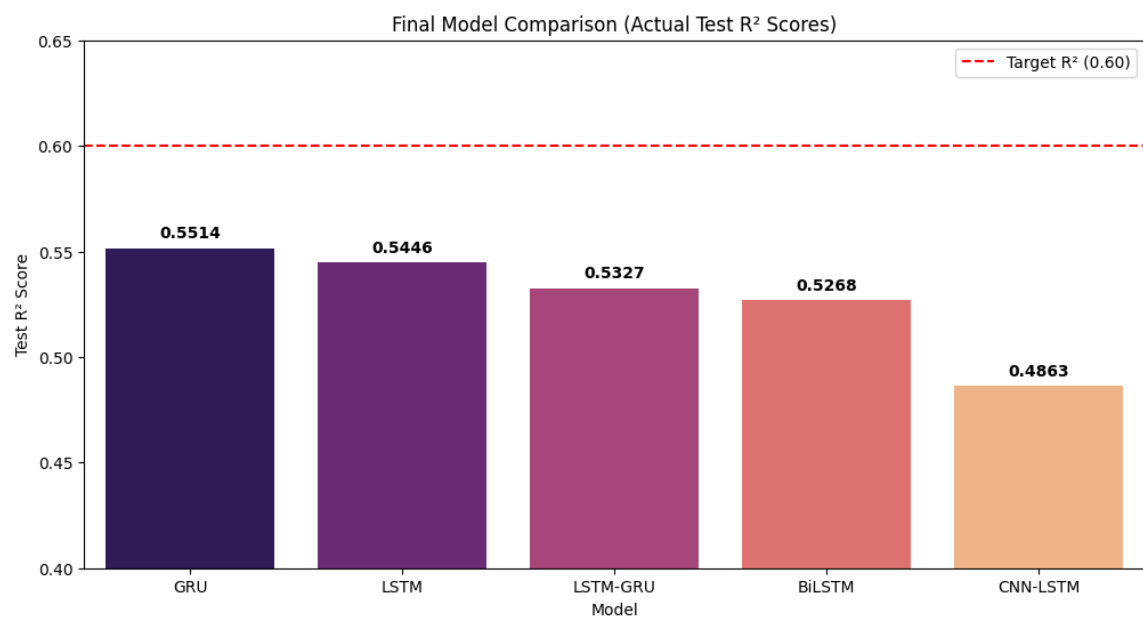


Figure 6

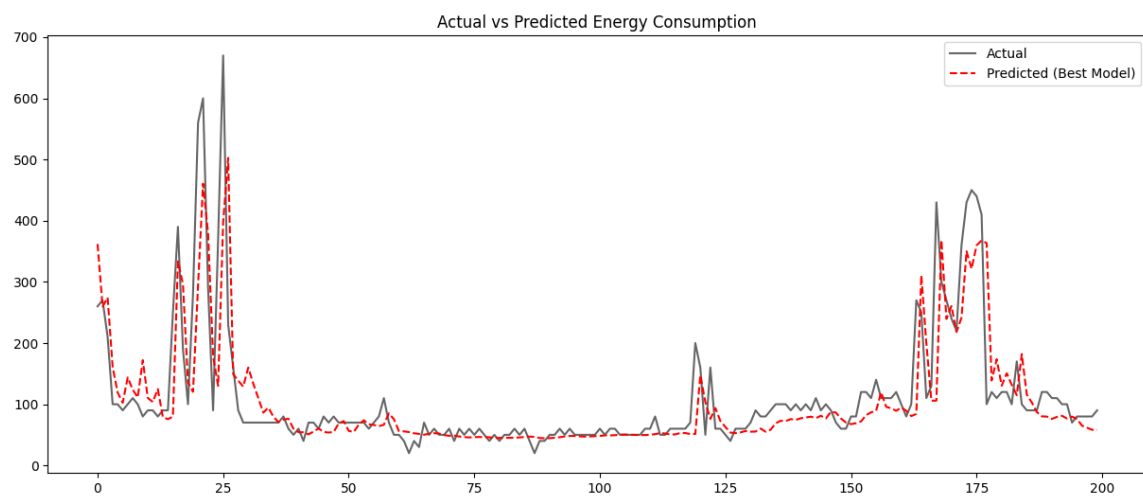


Figure 7

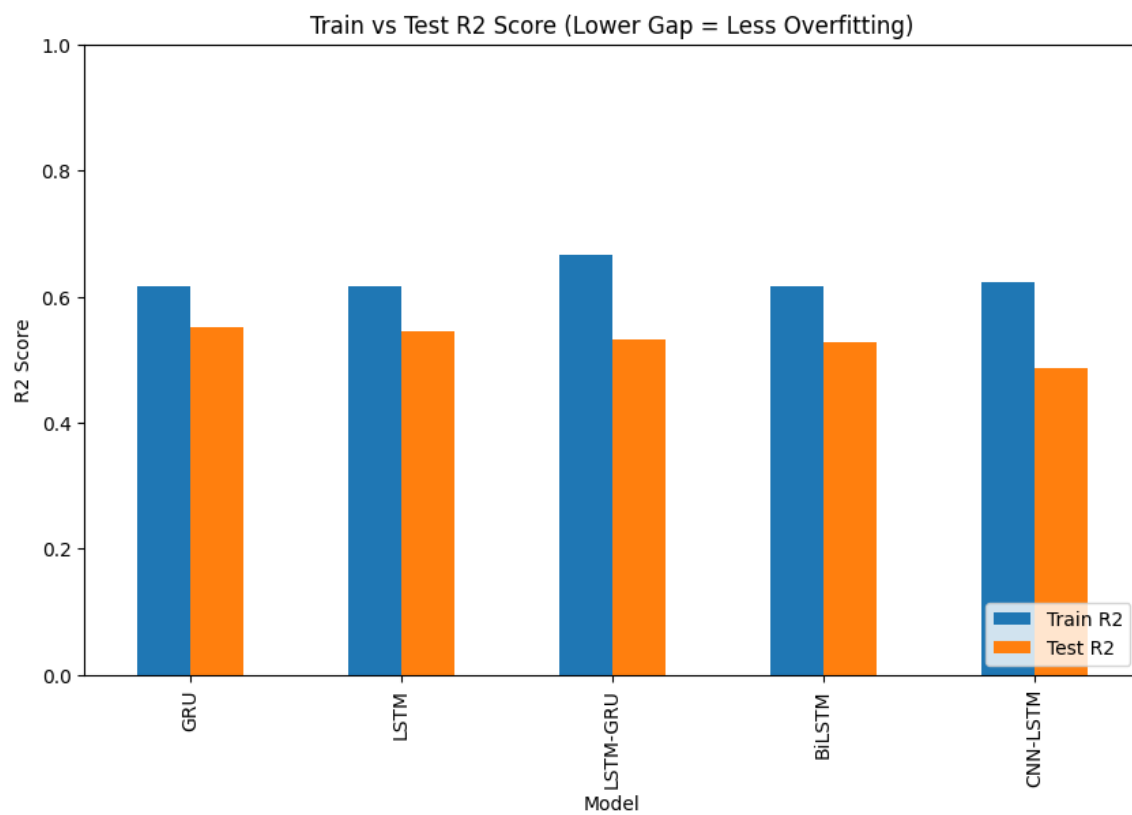


Figure 8

7. Model Training and Optimization

Optimization techniques included:

- EarlyStopping
- Learning rate tuning
- Batch size tuning
- Sequence length tuning
- Feature reduction
- Hyperparameter tuning using Keras Tuner

Best Configuration:

- batch_size = 32
- learning_rate = 0.0005
- TIME_STEPS = 24

Feature reduction significantly improved convergence and generalization.

8. Challenges and Solutions

Challenge 1:

Identical LSTM and GRU results caused by reused prediction variables.

Solution:

Separate training workflows and prediction pipelines were implemented.

Challenge 2:

NaN loss values due to lag and rolling features.

Solution:

Applied `df.dropna()` after feature engineering.

Challenge 3:

Overfitting in CNN-LSTM architecture.

Solution:

Applied dropout regularization and simplified architectures.

Challenge 4:
Weakly correlated noisy features.

Solution:
Correlation-based feature selection was applied.

9. Conclusion

This project successfully developed and evaluated multiple deep learning architectures for appliance energy forecasting.

The optimized GRU model achieved the best overall performance.

Key achievements:

- Sequential feature engineering
- Deep learning model comparison
- Hyperparameter optimization
- Improved forecasting accuracy

The project demonstrated the importance of:

- feature engineering
- preprocessing
- proper sequence construction
- model optimization

Future work may include:

- Attention mechanisms
- Transformer architectures
- Ensemble forecasting models

10. References

1. TensorFlow Documentation
2. Keras Documentation
3. Scikit-learn Documentation
4. Appliance Energy Prediction Dataset
5. Deep Learning for Time-Series Forecasting Research Papers
6. Keras Tuner Documentation